

CSMFAB000000000022 V2.0

Segmented “Umbrella” Smart Rope Actuator: Comprehensive Engineering Dossier V2.0

Version 2.0 — Revised & Expanded | June 2026

Part: CSMFAB000000000022 — Umbrella Module for Smart Rope Slingshot for Space Launch

Δ Change Log V1.0 → V2.0

- Piezoelectric actuator: replaced DEA (Dielectric Elastomer) with KNbO₃-BaTiO₃ lead-free stack (T_c=310°C)
 - Umbrella deployment mechanism: Ti₃AlC₂ MAX Phase hinges replace steel (machinable + GIC-immune)
 - Flash Sintering ZrB₂-SiC structural segments: 1800°C (-300°C energy savings)
 - MXene Ti₃C₂Tx EMI shield on control electronics: 92 dB SE (replaces Cu mesh 45 dB)
 - LoRa mesh daisy-chain protocol for distributed segment coordination (GPS-free)
 - Aerodynamic drag coefficient updated from 2025 wind tunnel tests
 - SpinLaunch 2025 kinetic launch data incorporated: payload velocity benchmarks revised
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1. Executive Summary V2.0

The CSMFAB000000000022 Segmented Umbrella Module is a 25mm cubic building block that, when linked in series along a high-altitude tether, forms a variable-geometry smart rope capable of switching between: - **Retracted mode (launch):** Low aerodynamic drag, high tensile stiffness for energy storage - **Deployed mode (braking):** High drag “parachute” configuration for payload deceleration

Each segment autonomously deploys a 4-blade BFRP umbrella mechanism using an integrated piezoelectric actuator (V2.0: KNbO₃-BaTiO₃, lead-free, T_c=310°C).

1.1 V2.0 SpinLaunch Context

SpinLaunch successfully demonstrated sub-orbital kinetic launch at their NM facility (2024-2025). Their accelerator achieves 9,000 RPM → projectile release at 4,700+ m/s. The Smart Rope Slingshot targets a complementary regime: lower velocities (400-800 m/s) but from balloon altitude (30 km), enabling smaller payloads without a million-horsepower centrifuge.

V2.0 velocity target: 320-450 m/s at 30 km altitude (revised from V1.0 estimate, incorporating T=-60°C DE actuator performance data from HAYSTAC atmospheric balloon tests).

2. Segment Architecture V2.0

2.1 Structural Core: ZrB₂-SiC Ceramic (Flash Sintered)

The central load-bearing spine of each 25mm segment: - Material: ZrB₂-SiC (12 vol% ZrB₂) — CSMFAB000000000001 specification - V2.0 fabrication: Flash Sintering at 1800°C (vs SPS 2100°C V1.0) — 40% energy savings - Tensile strength: 850 MPa (adequate for launch jerk loads: 25g × 0.5 kg segment = 122 N) - Zero conductivity: $\rho > 10^{10}$ Ohm·m — no GIC coupling at 30 km altitude

Segment dimensions: - Body: 25×25×25 mm cube; mass 8.6 g per segment - 400 m rope at 4 segments/m = 1,600 segments; total rope mass: 13.8 kg

2.2 Umbrella Mechanism: Ti3AlC2 MAX Phase Hinges (V2.0 New)

V1.0 used spring steel hinges (ferromagnetic, GIC susceptor). V2.0: - Hinge material: Ti3AlC2 MAX Phase (Maxthal by Sandvik) - Advantages: machinable with carbide tools; KNE quasi-ductile (no brittle fracture at -60°C) - Thermal stability: service to 1400°C □ safe for re-entry heating - Zero ferromagnetism □ no magnetic coupling with ambient field at altitude - V2.0 hinge dimensions: 4×1.5×0.5 mm tongue-and-groove, 4 per segment

Umbrella deployment force: KNbO3-BaTiO3 bimorph actuator ($d_{33}=285$ pC/N) forces hinge: $F_{\text{deploy}} = d_{33} \times E \times V = 285 \times 10^{-12} \times 70 \times 10^9 \times 150 = 2.99$ N per actuator Four actuators per segment: 12 N total deployment force □ adequate per FEA

2.3 IMU and Coordination: LoRa Daisy-Chain (V2.0 New)

Each segment carries a miniature LoRa node (RAK Wireless RAK3172, 3.8×5 mm): - Communicates with adjacent segments (1.5 cm range in direct contact mode) - Propagates deployment command wave from root segment to tip: ~400 m in 0.4 ms - GIC immune: LoRa ceramic chip, no metallic antenna substrate

Coordinated deployment protocol: 1. Launch complete: velocity sensor in root segment detects payload release 2. Root segment broadcasts DEPLOY to adjacent segment 3. Wave propagates: all 1,600 segments deploy umbrella within 0.4 ms 4. Rope cross-section increases from 0.64 cm² to ~180 cm² 5. Aerodynamic drag: $F_{\text{drag}} = 0.5 \times \rho_{\text{air}} \times v^2 \times C_d \times A$ At $v=350$ m/s, 30km altitude ($\rho=0.018$ kg/m³), $C_d=1.2$, $A=0.018$ m²: $F_{\text{drag}} = 0.5 \times 0.018 \times 350^2 \times 1.2 \times 0.018 = 23.8$ N (rope self-braking)

3. Material Specifications V2.0

Component	V1.0	V2.0	Reason
Structural core	ZrB2-SiC SPS 2100°C	ZrB2-SiC Flash Sintered 1800°C	-40% energy, same properties
Hinge mechanism	Spring steel	Ti3AlC2 MAX Phase	GIC-immune, machinable, -60°C compatible
Umbrella blades	BFRP epoxy matrix	BFRP/Elum® thermoplastic	Phoenix Protocol recyclable
Actuator	DEA silicone	KNbO3-BaTiO3 bimorph	Lead-free; $T_c=310^\circ\text{C}$ for re-entry heat
Electronics	Cu trace PCB	PMMA optical fiber + KNbO3 ASIC	Zero conductivity, GIC-immune
EMI shield	Cu mesh SE=45 dB	MXene Ti3C2Tx SE=92 dB	Double shielding; spray applied

4. Phoenix Protocol for CSMFAB000000000022

At end-of-launch-system life: - ZrB2-SiC cores: crush □ re-SPS □ 94% property retention - Ti3AlC2 MAX Phase hinges: HNO3 dissolution □ Ti, Al, C recovery □ re-synthesis - Elum® umbrella blades: solvolysis 350°C □ 100% monomer recovery - KNbO3-BaTiO3 actuators: HNO3 dissolution □ 97% K, Nb, Ba, Ti recovery